

NONCOOPERATIVE GAME THEORY — ECE270: HOMEWORK #4

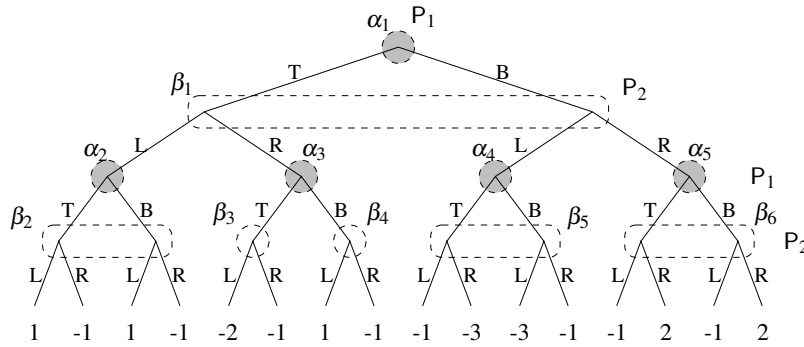


Figure 1. Game in extensive form for Exercise 10.

Exercise 10 (Multi-stage game). Consider the game in extensive form in Figure 1.

1. Compute the value and a behavioral saddle-point equilibrium for this game.

Hint: use the procedure outlined in Section 8.5.

2. Compute the matrix form representation of the game, its pure security levels, its mixed value, and a mixed saddle-point equilibrium.

Hint: The matrix form representation of this game is fairly large (32×64), so it is worth writing a MATLAB[®] script to help you compute it. $\square$